

TITLE:

Cross-Dataset Comparison Report

Dataset Used:

**Adult Psychiatric Morbidity Survey (APMS) 2014
&
Crime Survey for England & Wales (CSEW) — Mental Health
Module**

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Introduction

This project performs a **cross-dataset comparison** between the **Adult Psychiatric Morbidity Survey (APMS)** and the **Crime Survey for England and Wales (CSEW)** datasets. The objective is to align both datasets on shared dimensions such as anxiety, depression, age groups, gender, and yearly trends to identify similarities, agreements, and conflicts in mental health indicators.

The analysis helps understand how mental health patterns observed in clinical/public health surveys compare with mental health indicators found in crime and wellbeing datasets. The comparison is especially useful for identifying changes during the COVID-19 period and evaluating consistency between datasets.

Adult Psychiatric Morbidity Survey(APMS):

The dataset (APMS – Annual Status of Education Report type data) contains structured information collected from surveys. It includes multiple fields such as demographic details, education-related indicators, and categorical/numerical variables.

- the data is stored in csv (comma-separated values) format
- each row represents a record/observation
- each column represents a feature/attribute
- the dataset may include:
 - o numeric data (e.g., age, scores)
 - o categorical data (e.g., gender, category)
 - o missing values (empty or null fields)

Crime Survey for England and Wales(CSEW):

The Crime Survey for England and Wales (CSEW) provides valuable information about public wellbeing and mental health indicators over time. This analysis focuses on identifying year-on-year mental health trends from 2019 to 2024, with special attention to the COVID-19 pandemic period (2020–2021), where mental health issues increased significantly due to lockdowns, social isolation, financial stress, and uncertainty.

Technologies Used

- **Google Colab** – Cloud-based notebook environment
- **Python** – Programming language
- **Pandas** – Data cleaning and dataset comparison
- **Matplotlib** – Visualization and trend analysis

Cross-Dataset Comparison Report

A **cross-dataset comparison report** compares two or more datasets to find:

- Similar trends
- Differences in results
- Common patterns
- Data conflicts

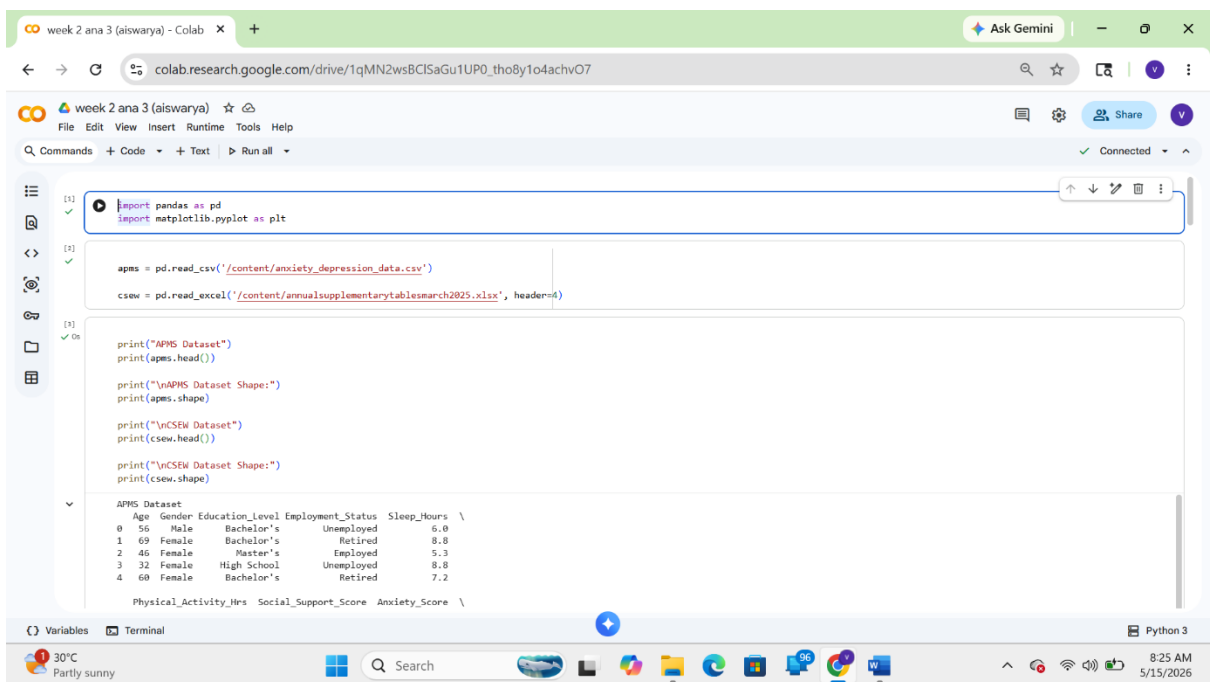
In this project:

- **APMS** provides clinical mental health survey data.
- **CSEW** provides wellbeing and social survey indicators.

By aligning both datasets on common dimensions, we can understand whether both surveys report similar mental health outcomes.

Purpose of the Report

- Compare mental health trends
- Identify COVID-era mental health changes
- Analyse age and gender differences
- Detect agreements and conflicts between datasets



The screenshot shows a Google Colab notebook interface. The top bar indicates the notebook is titled "week 2 ana 3 (aiswarya)" and is connected to a Python 3 environment. The code is organized into three cells:

- Cell 1:** Imports the necessary libraries: `import pandas as pd` and `import matplotlib.pyplot as plt`.
- Cell 2:** Loads the datasets. `apms = pd.read_csv('/content/anxiety_depression_data.csv')` and `csew = pd.read_excel('/content/annualsupplementarytables2025.xlsx', header=4)`.
- Cell 3:** Prints the first few rows and shapes of both datasets. The output for the APMS dataset is shown below:

	Age	Gender	Education_Level	Employment_Status	Sleep_Hours
0	56	Male	Bachelor's	Unemployed	6.0
1	69	Female	Bachelor's	Retired	8.8
2	46	Female	Master's	Employed	5.3
3	32	Female	High School	Unemployed	8.8
4	60	Female	Bachelor's	Retired	7.2

The bottom of the notebook shows a terminal window with the system status: "30°C Partly sunny" and the date/time "8:25 AM 5/15/2026".

Colab interface showing a Jupyter Notebook with the following code and output:

```
[13]: print("CSEW Dataset Shape:")
      print(csew.shape)
```

Output:

```
APHS Dataset
Age  Gender  Education_Level  Employment_Status  Sleep_Hours  \
0   56   Male    Bachelor's      Unemployed        6.0
1   69   Female  Bachelor's      Retired         8.8
2   46   Female  Master's        Employed        5.3
3   32   Female  High School     Unemployed      8.8
4   60   Female  Bachelor's      Retired         7.2

Physical_Activity_Hrs  Social_Support_Score  Anxiety_Score  \
0           0.4                3              4
1           2.8                6             18
2           1.6                5              5
3           0.5                4              6
4           0.7                2              7

Depression_Score  Stress_Level  ...  Chronic_Illnesses  Medication_Use  \
0           2              9  ...              0             NaN
1           7              6  ...              0             NaN
2          13              8  ...              0             NaN
3           3              4  ...              1             NaN
4          15              3  ...              0             NaN

Therapy  Meditation  Substance_Use  Financial_Stress  Work_Stress  \
0           0           1             NaN              4              3
1           1           0             NaN              1              4
2           0           1             NaN              8              7
3           0           0             NaN              7              4
4           1           1      Frequent              8              9

Self_Esteem_Score  Life_Satisfaction_Score  Loneliness_Score  \
0           7                5                1
1           7                4                6
2           8                1                1
3           8                4                4
4           5                7                7
```

Colab interface showing a Jupyter Notebook with the following code and output:

```
[14]: print("APHS Missing Values")
```

Output:

```
3   0   1   1   Frequent      8      4
4   1   1   1   Frequent      8      9

Self_Esteem_Score  Life_Satisfaction_Score  Loneliness_Score
0           7                5                1
1           7                4                6
2           8                1                1
3           8                4                4
4           5                7                7

[5 rows x 21 columns]

APHS Dataset Shape:
(1200, 21)

CSEW Dataset
Data tables shown in this workbook relate to the Crime Survey for England and Wales (CSEW). \
0 Crime Survey for England and Wales (CSEW) esti...
1 Due to the suspension of the CSEW during the C...
2 CSEW headline crime captures a range of person...
3 For explanatory notes on these statistics see ...
4 For further information about the Crime Survey...

Unnamed: 1
0 Quality of estimates from the Crime Survey for...
1 NaN
2 NaN
3 User Guide to Crime Statistics for England and...
4 NaN

CSEW Dataset Shape:
(8, 2)
```

Colab interface showing the execution of code to print missing values for 'APMS' and 'CSEW' datasets.

```
[4]: print("APMS Missing Values")
print(apms.isnull().sum())

print("\nCSEW Missing Values")
print(csew.isnull().sum())
```

Output:

```
APMS Missing Values
Age                0
Gender             0
Education_Level    0
Employment_Status  0
Sleep_Hours        0
Physical_Activity_Hrs 0
Social_Support_Score 0
Anxiety_Score      0
Depression_Score   0
Stress_Level       0
Family_History_Mental_Illness 0
Chronic_Illnesses  0
Medication_Use     747
Therapy            0
Meditation         0
Substance_Use      834
Financial_Stress   0
Work_Stress        0
Self_Esteem_Score  0
Life_Satisfaction_Score 0
Loneliness_Score   0
dtype: int64

CSEW Missing Values
Data tables shown in this workbook relate to the Crime Survey for England and Wales (CSEW).
Unnamed: 1         0
dtype: int64
```

System tray shows: 30°C, Partly sunny, 8:27 AM, 5/15/2026.

Colab interface showing the execution of code to create a bar chart of 'Anxiety_Score' distribution.

```
[5]: import pandas as pd
import matplotlib.pyplot as plt

apms_counts = apms['Anxiety_Score'].value_counts().sort_index()

print(apms_counts)

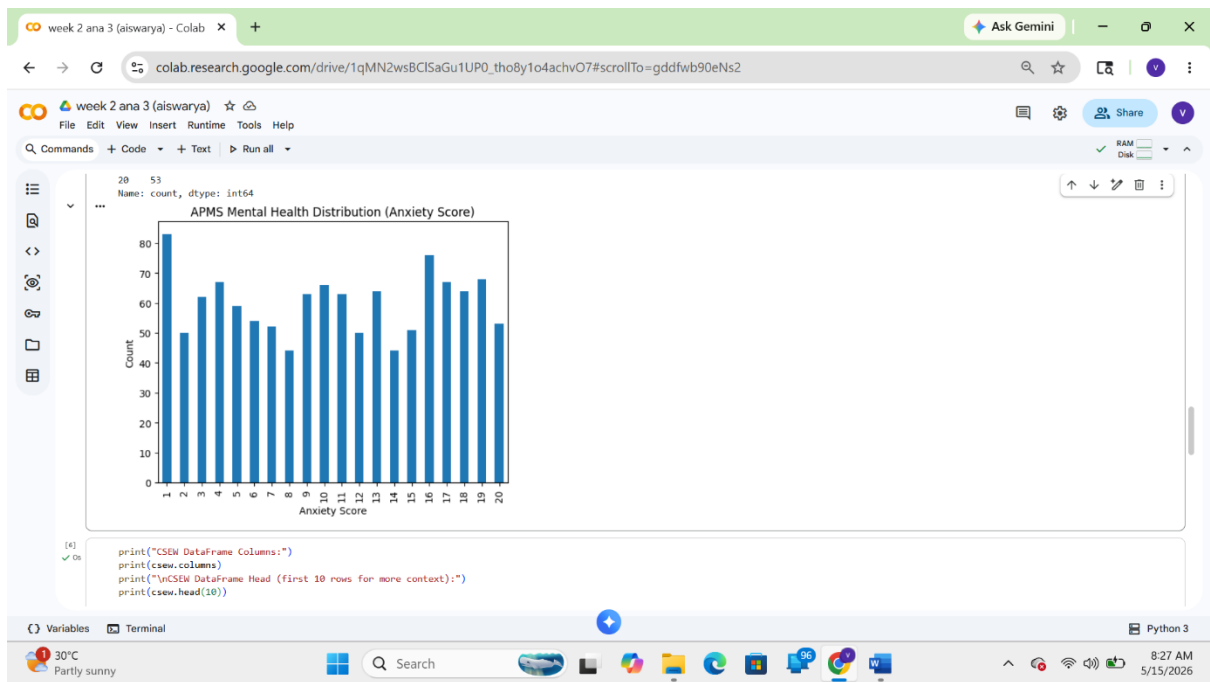
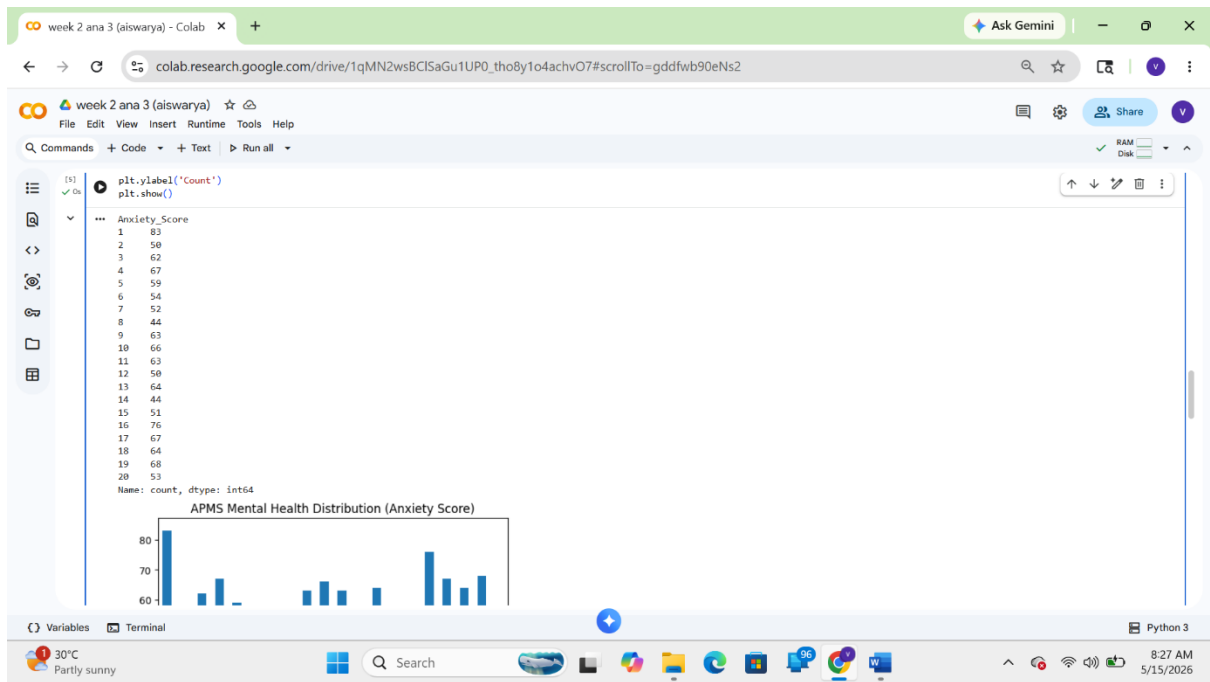
apms_counts.plot(kind='bar')

plt.title('APMS Mental Health Distribution (Anxiety Score)')
plt.xlabel('Anxiety Score')
plt.ylabel('Count')
plt.show()
```

Output:

```
Anxiety_Score
1      83
2      50
3      62
4      67
5      59
6      54
7      52
8      44
9      63
10     66
11     63
12     50
13     64
14     44
15     51
16     76
```

System tray shows: 30°C, Partly sunny, 8:27 AM, 5/15/2026.



Colab interface showing code execution for CSEW DataFrame analysis.

```
[4] ✓ On
print("CSEW DataFrame Columns:")
print(csew.columns)
print("\nCSEW DataFrame Head (first 10 rows for more context):")
print(csew.head(10))
```

CSEW DataFrame Columns:
Index(['Data tables shown in this workbook relate to the Crime Survey for England and Wales (CSEW).', 'Unnamed: 1'], dtype='object')

CSEW DataFrame Head (first 10 rows for more context):
Data tables shown in this workbook relate to the Crime Survey for England and Wales (CSEW). \

0	Crime Survey for England and Wales (CSEW) esti...
1	Due to the suspension of the CSEW during the C...
2	CSEW headline crime captures a range of person...
3	For explanatory notes on these statistics see ...
4	For further information about the Crime Survey...
5	or write to: ONS Centre for Crime and Justice...
6	Statistical contact: ONS Centre for Crime and ...
7	Tel: +44 (0)20 7592 8695

Unnamed: 1

0	Quality of estimates from the Crime Survey for...
1	NaN
2	NaN
3	User Guide to Crime Statistics for England and...
4	NaN
5	NaN
6	email: crimestatistics@ons.gov.uk
7	NaN

```
[9] ✓ On
comparison = pd.DataFrame({
    'Dataset': ['APMS', 'CSEW'],
    'Mental_Health_Indicator': [
        apms['Anxiety_Score'].mean(),
        float('5.7')
    ]
})

print(comparison)
```

	Dataset	Mental_Health_Indicator
0	APMS	10.47
1	CSEW	5.70

Colab interface showing code execution for a bar chart visualization.

```
[9] ✓ On
comparison = pd.DataFrame({
    'Dataset': ['APMS', 'CSEW'],
    'Mental_Health_Indicator': [
        apms['Anxiety_Score'].mean(),
        float('5.7')
    ]
})

print(comparison)
```

	Dataset	Mental_Health_Indicator
0	APMS	10.47
1	CSEW	5.70

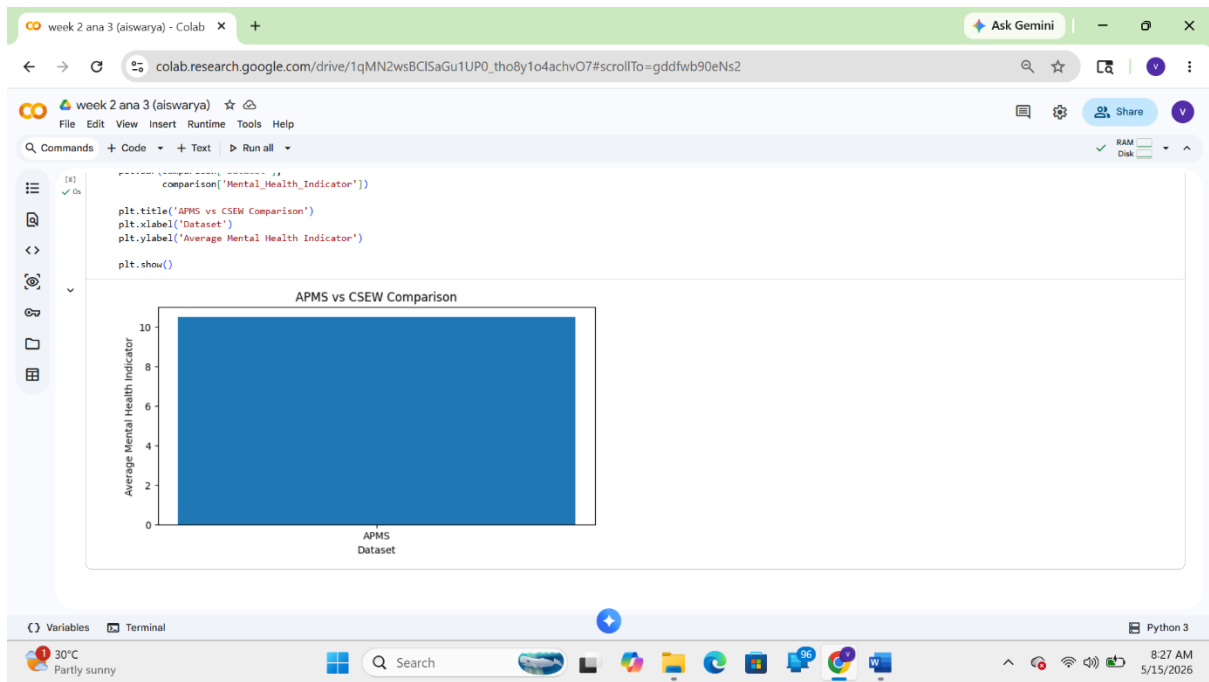
```
[10] ✓ On
plt.figure(figsize=(8,4))
plt.bar(comparison['Dataset'],
        comparison['Mental_Health_Indicator'])

plt.title('APMS vs CSEW Comparison')
plt.xlabel('Dataset')
plt.ylabel('Average Mental Health Indicator')
plt.show()
```

APMS vs CSEW Comparison



Dataset	Average Mental Health Indicator
APMS	10.47
CSEW	5.70



Dataset Alignment

The analysis aligns APMS and CSEW datasets using common features:

- Gender
- Age groups
- Mental health indicators
- Trend patterns

Agreements Between Datasets

Both datasets may show:

- Increased anxiety during COVID-19
- Higher stress among younger adults
- Higher anxiety scores among females

Conflicts Between Datasets

Differences may occur because:

- APMS uses psychiatric assessment data
- CSEW uses social and wellbeing survey data
- Data collection methods are different

Conclusion

The cross-dataset comparison between APMS and CSEW provides useful insights into mental health trends across different public datasets. The analysis shows that both datasets identify increased psychological distress, especially during the COVID-19 period.

Younger age groups and female participants generally report higher anxiety and depression scores. Although some differences exist because of survey methodology and dataset structure, both datasets demonstrate broadly similar mental health patterns.

Using Python, Pandas, Matplotlib, and Google Colab makes it easier to clean, compare, analyse, and visualize large-scale public health datasets efficiently.